

14

Audio

If you've ever watched a horror film with your speakers muted, you know just how important audio is to immersiveness. (If not, try it! It's a real ear-opener.) Be it a film or a video game, sound can quite literally make the difference between a gripping, emotional, unforgettable multimedia experience and a lackluster yawnfest.

Modern games immerse the player in a realistic (or a semi-realistic but stylized) virtual environment. The graphics engine is charged with the task of reproducing as accurately and believably as possible what the player would actually see, if he or she were present within this virtual environment (while remaining true to the art style of the game). In exactly the same sense, the audio engine is charged with the task of accurately and believably reproducing what the player would actually hear, if he or she were present in the game world (while remaining true to the fiction and tonal style of the game). Sound programmers today use the term audio *rendering* engine to underscore its many parallels with the graphics rendering engine.

In this chapter, we're going to explore both the theory and practice of creating audio for a triple-A game. We'll introduce an area of mathematics called *signal processing theory* that underlies almost every aspect of digital audio technology, including digital sound recording and playback, filtering, reverb and other digital signal processor (DSP) effects. We'll explore game audio from a software engineering standpoint, by investigating a number of widely used

audio APIs, breaking down the components that comprise a typical audio rendering engine and learning how the audio system is interconnected with other game engine systems. We'll also see how environmental acoustic modeling and character dialog were handled in Naughty Dog's popular game *The Last of Us*. So hold on tight, keep your hands inside the car at all times and enjoy the noisy ride!

14.1 The Physics of Sound

Sound is a compression wave that travels through the air (or some other compressible medium). A sound wave gives rise to alternating regions of air compression and decompression (also known as *rarefaction*) relative to the average atmospheric pressure. As such, we measure the *amplitude* of a sound wave in units of *pressure*. In SI units, pressure is measured in Pascals, abbreviated Pa. One Pascal is the force of one Newton applied over an area of one square meter ($1 \text{ Pa} = 1 \text{ N/m}^2 = 1 \text{ kg}/(\text{m} \cdot \text{s}^2)$).

The *instantaneous acoustic pressure* is the ambient atmospheric pressure (considered a constant for our purposes) plus the perturbation caused by the sound wave at one specific instant in time:

$$p_{\text{inst}} = p_{\text{atmos}} + p_{\text{sound}}.$$

Of course, sound is a dynamic phenomenon—the sound pressure varies over time. We can plot the instantaneous sound pressure as a function of time, $p_{\text{inst}}(t)$. In *signal processing theory*—the area of mathematics that underlies virtually every aspect of digital audio technology—such a time-varying function is called a *signal*. Figure 14.1 illustrates a typical sound wave signal $p(t)$, oscillating about the average atmospheric pressure.

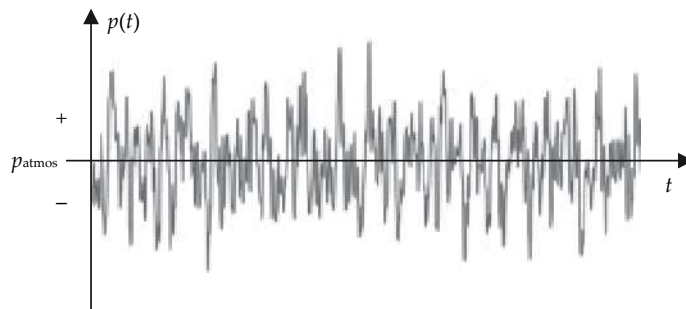


Figure 14.1. A signal $p(t)$ can be used to model the time-varying acoustic pressure of a sound.

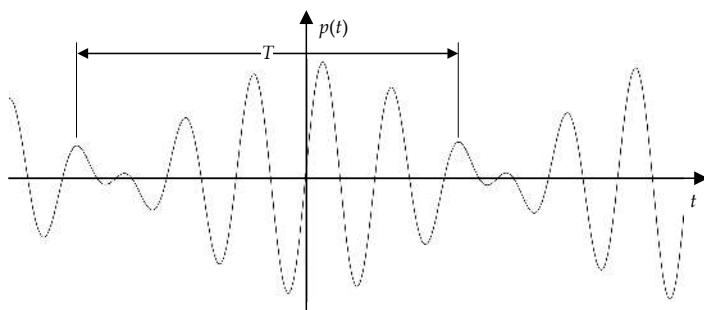


Figure 14.2. The period T of an arbitrary periodic signal is the minimum time between repeated patterns in the waveform.

14.1.1 Properties of Sound Waves

When a musical instrument plays a long steady note, the resulting sound pressure signal is *periodic*, meaning the waveform consists of a *repeating pattern* characteristic to that particular kind of instrument. The *period* T of any repeating pattern describes the minimum amount of time that passes between successive instances of the pattern. For example, for a sinusoidal sound wave the period measures the time between successive peaks or troughs. In SI units, period is typically measured in seconds (s). This is illustrated in Figure 14.2.

The *frequency* of a wave is just the inverse of its period ($f = 1/T$). Frequency is measured in Hertz (Hz), which means “cycles per second.” A “cycle” is technically a dimensionless quantity, so the Hertz is the inverse of the second ($\text{Hz} = 1/\text{s}$).

Many scientists and mathematicians make use of a quantity known as the *angular frequency*, typically denoted by the symbol ω . The angular frequency is just the rate of oscillation measured in *radians* per second instead of *cycles* per second. Since one complete circular rotation is 2π radians, $\omega = 2\pi f = 2\pi/T$. Angular frequency is very useful when analyzing sinusoidal waves, because a circular motion in two dimensions gives rise to a sinusoidal motion when projected onto a single dimensional axis.

The amount by which a periodic signal such as a sine wave is *shifted* left or right along the time axis is known as its *phase*. Phase is a relative term. For example, $\sin(t)$ is really just a version of $\cos(t)$ that has been *phase-shifted* by $+\frac{\pi}{2}$ along the t axis (i.e., $\sin(t) = \cos(t - \frac{\pi}{2})$). Likewise $\cos(t)$ is just $\sin(t)$ phase-shifted by $-\frac{\pi}{2}$ (i.e., $\cos(t) = \sin(t + \frac{\pi}{2})$). Phase is illustrated in Figure 14.3.

The *speed* v at which a sound wave propagates through its medium depends upon the material and physical properties of the medium, including phase (solid, gas or liquid), temperature, pressure and density. In 20°C dry

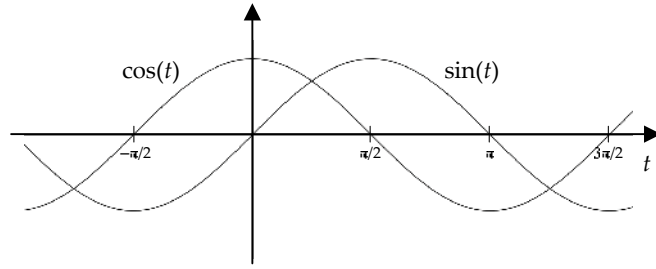


Figure 14.3. The sine and cosine functions are just phase-shifted versions of one another.

air, the speed of sound is approximately 343.2 m/s, which is 767.7 mph or 1235.6 km/h.

The *wavelength* λ of a sinusoidal wave measures the spatial distance between successive peaks or troughs. It depends in part on the frequency of the wave, but because it is a *spatial* measurement it also depends on the speed of the wave. Specifically, $\lambda = v/f$ where v is the speed of the wave (measured in m/s) and f is the frequency (measured in Hz or 1/s). The seconds in the numerator and denominator cancel one another, and we are left with wavelength measured in units of meters.

14.1.2 Perceived Loudness and the Decibel

In order to judge the “loudness” of the sounds we hear, our ears continuously *average* the amplitude of the incoming sound signal over a short, sliding time window. This averaging effect is modeled well by a quantity known as the *effective sound pressure*. This is defined as the *root mean square* (RMS) of the instantaneous sound pressure measured over a specific interval of time.

If we were to take a series of n discrete sound pressure measurements p_i , equally spaced in time, the RMS sound pressure p_{rms} would be

$$p_{\text{rms}} = \sqrt{\frac{1}{n} \sum_{i=1}^n p_i^2}. \quad (14.1)$$

However, our ears take pressure measurements continuously, rather than at discrete points in time. If we imagine measuring the instantaneous sound pressure continuously, starting at time T_1 and lasting until time T_2 , the summation in Equation (14.1) would become an integral as follows:

$$p_{\text{rms}} = \sqrt{\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} (p(t))^2 dt}. \quad (14.2)$$

However, the story doesn't end here. Perceived loudness is actually proportional to the *acoustic intensity* I , which is itself proportional to the *square* of the RMS sound pressure:

$$I \propto p_{\text{rms}}^2.$$

Humans can perceive a very wide range of sound pressure variations—from the flutter of a piece of paper falling to the ground to the boom of an aircraft breaking mach one. In order to manage such a wide dynamic range, we normally measure sound intensity in units of *decibels* (dB). The decibel is a *logarithmic* unit that expresses the *ratio* between two values. By employing a logarithmic scale, the decibel allows a wide range of measurements to be represented by a relatively narrow range of values. A decibel is actually one-tenth of a *bel*, a unit named in honor of Alexander Graham Bell.

When sound intensity is measured in decibels, it is called *sound pressure level* (SPL) and represented by the symbol L_p . Sound pressure level is defined as the ratio of the acoustic intensity (i.e., the squared pressure) of a sound relative to a reference intensity p_{ref} that represents the lower limit of human hearing. So we have:

$$\begin{aligned} L_p &= 10 \log_{10} \left(\frac{p_{\text{rms}}^2}{p_{\text{ref}}^2} \right) \text{ dB} \\ &= 20 \log_{10} \left(\frac{p_{\text{rms}}}{p_{\text{ref}}} \right) \text{ dB}, \end{aligned}$$

where the 20 arises because when we take the square outside the logarithm, it becomes a multiplication by two. The commonly used reference sound pressure in air is $p_{\text{ref}} = 20 \mu\text{Pa}$ (RMS). For more information on sound pressure, the physics of sound and human aural perception, see [6].

By the way, if you're feeling a bit rusty, the following identities may help to refresh your memory on logarithms. In Equations (14.3), b , x and y are positive real numbers with $b \neq 1$, c and d are any real numbers, $c = \log_b x$, and $d = \log_b y$ (or written another way, $b^c = x$ and $b^d = y$).

$\log_b x = c$	when	$b^c = x$	(definition);
$\log_b 1 = 0$	because	$b^0 = 1$;	
$\log_b b = 1$	because	$b^1 = b$;	
$\log_b(x \cdot y) = \log_b x + \log_b y$	because	$b^c \cdot b^d = b^{c+d}$;	(14.3)
$\log_b(x/y) = \log_b x - \log_b y$	because	$b^c / b^d = b^{c-d}$;	
$\log_b x^d = d \log_b x$	because	$(b^c)^d = b^{cd}$.	

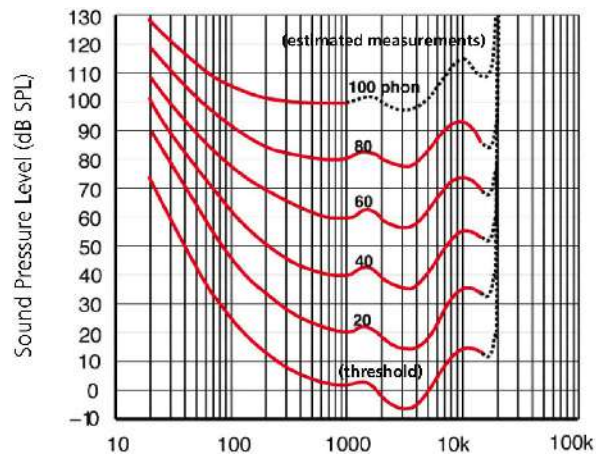


Figure 14.4. The human ear is most sensitive in the frequency range between 2 and 5 kHz. As the frequency decreases or increases beyond this range, more and more acoustic pressure is required to produce the same perception of “loudness.”

14.1.2.1 Equal-Loudness Contours

The human ear does not have the same response to sound waves of different frequencies. The human ear is most sensitive in the frequency range between 2 and 5 kHz. As the frequency decreases or increases beyond this range, more and more acoustic intensity (i.e., pressure) is required to produce the same perception of “loudness.”

Figure 14.4 shows a number of equal-loudness countours, each corresponding to a different perceived loudness level. These curves show that more pressure is required at low and high frequencies to achieve the same perceived loudness than is needed at mid-range frequencies. Or put another way, if we were to keep the amplitude of an acoustic pressure wave the same while varying the frequency, the human ear would actually perceive the lower and higher frequencies as “less loud” than the mid-range frequencies. The lowest equal-loudness contour represents the quietest audible tone and is also known as the absolute threshold of hearing. The highest contour passes through the human threshold of pain, which lies roughly at the 120 dB level for audible sounds.

For more information on equal-loudness contours, and the Fletcher-Munson curves on which they are based, see <https://bit.ly/2HfCjCs>.

14.1.2.2 The Audible Frequency Band

A typical adult can hear sounds with frequencies as low as 20 Hz and as high as 20,000 Hz (20 kHz) (although the upper limit generally decreases with age). The equal-loudness contours help to explain why the human ear can perceive sounds only within this limited “band” of frequencies. As the frequency becomes lower or higher, more and more acoustic pressure is required to produce the same perceived loudness. As the frequency approaches the lower or upper limits of human hearing, the contours become asymptotically vertical, meaning we’d need an effectively infinite acoustic pressure to produce any perception of loudness at all. Or put another way, human audio perception drops off to effectively zero outside the audible frequency band.

14.1.3 Sound Wave Propagation

Like any kind of wave, an acoustic pressure wave propagates through space and can be *absorbed* or *reflected* by surfaces, *diffracted* around corners and through narrow “slits,” and *refracted* as it passes across the boundary between different transmission media. Sound waves exhibit no *polarization*¹ because the acoustic pressure oscillation occurs in the direction of wave travel (this is known as a *longitudinal* wave), rather than perpendicular to it as with light waves (a *transverse* wave). In games, we typically model the absorption, reflection and sometimes the diffraction (e.g., bending slightly around corners) of our virtual sound waves, but we generally ignore refraction effects because these effects are not easily noticed by a human listener.

14.1.3.1 Fall-Off with Distance

In an open space with otherwise perfectly still air, and assuming a sound source that radiates equally in all directions, the intensity of the sound pressure wave it produces falls off with distance, following a $1/r^2$ law, while pressure follows a $1/r$ law.

$$p(r) \propto \frac{1}{r};$$

$$I(r) \propto \frac{1}{r^2}.$$

Here, r measures the radial distance of the listener or microphone from the sound source, and both pressure and intensity are expressed as functions of r .

¹Sound waves in solids can be transverse and therefore can exhibit polarization.

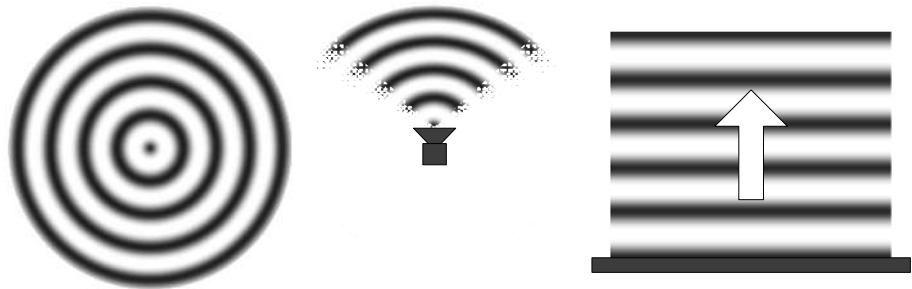


Figure 14.5. Three types of sound sources and their sound radiation patterns (in two dimensions for ease of illustration). From left to right: omnidirectional, conical and directional.

More precisely, the sound pressure level for a spherically radiating (omnidirectional) sound wave in open space can be written as follows:

$$\begin{aligned} L_p(r) &= L_p(0) + 10 \log_{10} \left(\frac{1}{4\pi r^2} \right) \text{ dB} \\ &= L_p(0) - 10 \log_{10} (4\pi r^2) \text{ dB}, \end{aligned}$$

where $L_p(r)$ is the SPL at the listener as a function of its radial distance from the sound source, and $L_p(0)$ represents the unattenuated or “natural” sound intensity of the source.

Sound sources are not always omnidirectional. For example, when a large, flat wall reflects sound waves, it acts like a purely *directional* sound source—the reflected waves propagate in a single direction, and the pressure wavefronts are essentially parallel.

A bullhorn projects sound in a particular direction but with a *conical* fall-off, meaning that the intensity of the sound waves is maximum along the centerline of the projection “cone,” but falls off as the angle between the listener and this centerline increases.

Various sound radiation patterns are illustrated in Figure 14.5.

14.1.3.2 Atmospheric Absorption

The $1/r$ fall-off of sound pressure with distance arises because energy is dissipated as the waveform expands geometrically. This fall-off affects sounds of all frequencies equally. Sound intensity also falls off with distance due to energy absorption by the atmosphere. Atmospheric absorption effects are *not* uniform across the entire frequency spectrum. In general, the absorption effect becomes greater as the frequency of the sound increases.

I’m reminded of a story I heard when I was in high school: A woman was walking down a quiet village street at night. She heard a sporadic sequence of

low tones with long, silent gaps between them. Curious what might be making these strange tones, she walked toward them. As she walked, the tones became louder and the spaces between the tones seemed to get shorter. After a few minutes' walk, the tones had resolved into a beautiful piece of music. The woman arrived at an open window to discover a viola player practicing within. The musician stopped playing to say "hello," and the woman asked him why he had been playing random notes a few minutes before. He replied, "I haven't been playing random notes—I've been playing this piece the whole time." The explanation for what the woman heard, of course, is that lower-frequency sounds can be heard over longer distances than higher-frequency sounds due to atmospheric absorption. You can learn more about sound wave propagation at http://www.sfu.ca/sonic-studio/handbook/Sound_Propagation.html.

Other factors also affect the intensity of sound waves as they propagate through their medium. In general, fall-off depends on distance, frequency, temperature and humidity. See <http://sengpielaudio.com/calculator-air.htm> for an online calculator that lets you experiment with the effects of these factors.

14.1.3.3 Phase Shift and Interference

When multiple sound waves overlap in space, their amplitudes add together—this is called *superposition*. Consider two periodic sound waves with the same frequency. (The simplest example would be two sinusoids.) If the waves are *in phase*—that is, their peaks and troughs line up—then the waves will *positively reinforce* each other, and the result is a wave with larger amplitude than either of the original waves. Likewise, if the waves are *out of phase*, the peaks of one wave can tend to *cancel* the troughs of the other and vice versa, and the result is a wave with lower (or even zero) amplitude.

When multiple waves interact, we call this *interference*. *Constructive interference* describes the case in which the waves reinforce one another and the amplitude increases. *Destructive interference* occurs when the waves cancel each other out, resulting in lower amplitude.

The frequency of the waves has an important effect on this phenomenon: If the frequencies of the two waves match closely, the interference simply increases or decreases the overall amplitude. If the frequencies differ significantly, we can get an effect called *beating*, wherein the frequency difference causes alternating periods of the waves being in and out of phase, resulting in alternating periods of higher and lower amplitude.

Interference can occur between two totally unrelated sound signals, or it can occur if a single sound signal takes multiple paths from the source to the

listener. In the latter case, the difference in path lengths introduces a phase shift that can cause either constructive or destructive interference, depending on the amount of the phase shift.

Comb Filtering

Interference can lead to an effect known as *comb filtering*. This is caused when sound waves reflect off surfaces in such a way as to either almost completely cancel or completely reinforce certain frequencies. The result is a *frequency response* (see Section 14.2.5.7) with lots of narrow peaks and troughs, which when plotted look a bit like a comb (hence the name). This effect can have a big impact on audio reproduction and recording—sometimes it is an undesirable artifact, and sometimes it is used as a tool. The existence of comb filtering is also one of the key reasons why it is generally better to spend money on acoustic room treatment than to spend money on high-end audio equipment: If the room exhibits comb effects, you’re wasting your time trying to get a flat response from your gear. See http://www.realtraps.com/video_comb.htm for a great video by Ethan Winer on the subject.

14.1.3.4 Reverb and Echo

In any environment containing sound-reflective surfaces, a listener generally receives three kinds of sound waves from a sound source:

- *Direct (dry)*. Sound waves that arrive at the listener via a direct, unobstructed path from the source are collectively known as *direct* or *dry* sound.
- *Early reflections (echo)*. Sound waves that arrive at the listener via an indirect path, after being reflected from and partially absorbed by surrounding surfaces, take a longer time to reach the listener because their path is longer. As such, there will be a *delay* between the arrival of the direct sound waves and the arrival of the reflected waves. The first group of reflected sound waves arriving at the ear have only interacted with one or two surfaces. As such, they are relatively “clean” signals, and we perceive them as distinct new “copies” of the sound or *echos*.
- *Late reverberations (tail)*. Once the sound waves have bounced around the listening space more than a few times, they superimpose and interfere with one another so much that the brain can no longer detect distinct echos. These are known as *late reverberations* or the *diffuse tail*. The properties of the reflective surfaces cause the amplitudes of the waves to be attenuated by varying amounts. And because the reflected sound waves

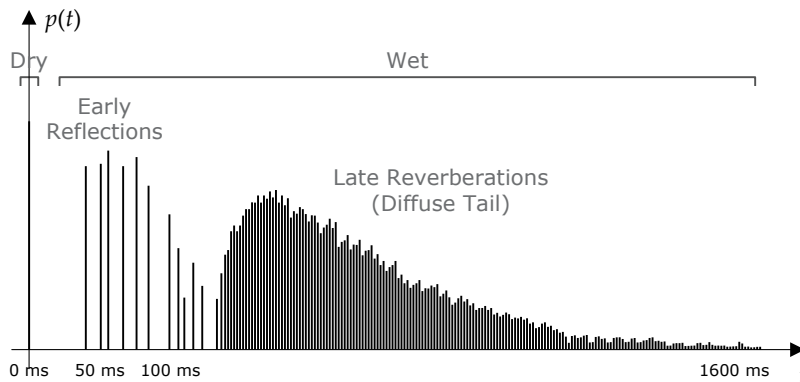


Figure 14.6. Direct sound waves, early reflections and late reverberations.

are delayed, phase shifts occur causing the waves to interfere with one another. This causes certain frequencies to be attenuated relative to the others. When we speak of the *acoustics* of a space, we are largely speaking about the effects of late reverberations on the perceived “quality” or “timbre” of the sound.

Collectively, the echos and the tail combine with the dry sound to create what is known as *wet* sound. Figure 14.6 illustrates the wet and dry components of a single abrupt clap.

The early reflections and late reverberations provide the brain with a wealth of cues that tell us quite a lot about the type of space we’re in. The *pre-delay* is the time interval between the arrival of the direct sound waves and the arrival of the very first reflected waves. From the pre-delay, the brain can determine the approximate size of the room or space in which we are listening. The *decay* is the time it takes for the reflected sound waves to die away. This tells our brains how much of the sound has been absorbed by the surroundings, and so indirectly tells us something about the materials that make up the space we’re in. For example, a small tiled bathroom would produce late reverberations with a very short pre-delay (due to its small size) and a long decay (due to the tile’s ability to reflect sound waves efficiently, with little absorption). A large granite-walled room like Grand Central Terminal (a.k.a. Grand Central Station) in New York City will have a much longer pre-delay and a lot more echos, but the decay will be similar to that of the tiled bathroom.

If we were to hang curtains in that bathroom, or if the walls were covered with wood panels instead of tile, the pre-delay would remain the same, but

the decay, along with other factors such as *density* (how closely spaced in time the individual reflections are) and *diffusion* (the rate at which the reflections increase in density over time), would change. This explains how a person can guess where they are even when blindfolded, or how the blind can learn to navigate with only a cane to aid them. Sound provides us with a *lot* of information about our surroundings!

The term *reverb* is used to describe the quality of a sound in terms of its wet components. In the early days of audio recording, sound engineers had little control over reverb, relying entirely on the shape and construction of the room in which the recording was made. Later, simple artificial reverb devices were created, from the use of a speaker and microphone in a bathroom by Bill Putman Sr. (founder of Universal Audio), to the use of a long metal plate or spring to introduce a delay in a sound signal, to modern digital techniques. Today, digital signal processor (DSP) chips and/or software are used not only to recreate natural reverb effects in recorded sound effects and music, but also to augment recordings with all sorts of interesting effects that are not normally heard in nature. We'll learn more about digital signal processing in Section 14.2. You can read more about reverb at <http://www.uaudio.com/blog/the-basics-of-reverb>.

An *anechoic chamber* is a room especially designed to entirely eliminate reflected sound waves. This is accomplished by lining the walls, floor and ceiling of the room with thick corrugated foam padding that absorbs essentially all of the reflected sound waves. As a result, only the direct (dry) sound reaches the listener or microphone. Sound in an anechoic chamber has a completely "dead" timbre. Anechoic chambers are useful for recording "pure" sounds that contain no reverb. Such pure sounds are often perfect candidates for input into a digital signal processing pipeline, giving a sound designer maximum flexibility to control the timbre of the sound.

14.1.3.5 Sound in Motion: The Doppler Effect

If you've ever stood at a railway crossing when a train goes by, you've heard the *Doppler effect* in action. The sound of the train seems higher pitched when it is approaching you, and becomes lower pitched as it races off into the distance. Sound waves travel at a roughly constant speed through the air, but the sound source (in this case, the train) is also moving. The sound waves that are moving in the same direction as the train become "squashed together," and the waves that are moving opposite to the motion of the train become "spread out," each by an amount proportional to the difference between the speed of sound in air and the speed of the train through the air. The frequency of the squashed waves is therefore increased, because the space between the peaks and troughs

of the sound waves has been effectively reduced, resulting in a higher-pitched sound. Likewise, the frequency of the spread-out waves is decreased, resulting in a lower-pitched sound. The Doppler effect was named after the Austrian physicist Christian Doppler, who identified it in 1842.

The Doppler effect also occurs when the listener is moving but the sound source is stationary. In general, the Doppler shift is dependent upon the *relative velocity* (as a vector) between the listener and the sound source. In one dimension, the Doppler shift amounts to a change in frequency, and can be quantified as follows:

$$f' = \left(\frac{c + v_l}{c + v_s} \right) f,$$

where f is the original frequency, f' is the Doppler-shifted (observed) frequency at the listener, c is the speed of sound in air and v_l and v_s are the speeds of the listener and sound source, respectively. If the speeds of the sound sources are very small relative to the speed of sound, we can approximate this relationship as follows:

$$\begin{aligned} f' &= \left(\frac{1 + (v_l - v_s)/c}{1} \right) f \\ &= \left(\frac{1 + \Delta v/c}{1} \right) f. \end{aligned}$$

This expression makes the relative velocity, Δv , apparent. The Doppler effect can be easily visualized by looking at the following animated GIF: <http://en.wikipedia.org/wiki/File:Dopplereffectsourcemovingrightatmach0.7.gif>.

14.1.4 Perception of Position

The human auditory system has evolved to allow a reasonably accurate perception of the position of sounds in the space around us. A number of factors contribute to our perception of sound position:

- *Fall-off with distance* provides us with a rough idea of how far away the source of a sound is. In order for this to work, we must have some idea of the loudness of the sound when heard at close range to serve as a “baseline.”
- *Atmospheric absorption* causes the higher frequencies in a sound to drop out as the source moves farther away from the listener. This can serve as an important cue in perceiving the difference between, for example, a person speaking at normal volume but far away and a person speaking at reduced volume up close.